

Applications of Neuromorphic/Event Camera in Robotics With Human in Loop: A Systematic Review, Datasets, and Challenges

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Abstract—The evolution of industrial robotics has advanced from isolated, caged systems through basic human–robot interaction (HRI) to sophisticated human–robot collaboration (HRC). However, conventional vision systems based on red, green, blue (RGB) cameras remain a significant limiting factor in realizing the full potential of collaborative automation. This comprehensive review examines the transformative role of event cameras in advancing HRC capabilities and addressing current limitations in industrial settings. Event cameras, with their microsecond-level temporal resolution and robust performance under challenging lighting conditions, offer substantial advantages over traditional RGB cameras that are constrained by fixed frame rates and ambient lighting dependencies. We present a systematic framework for leveraging event cameras to enhance the human state understanding in collaborative robotics, encompassing real-time detection of poses, gestures, facial expressions, and emotional states. This framework addresses fundamental challenges in workplace safety and collaborative efficiency while enabling more sophisticated and responsive HRC systems. Our review synthesizes recent research developments in event camera applications specific to HRC, providing a detailed comparative analysis of their advantages over conventional vision systems. We identify emerging opportunities and potential research directions for advancing event-based vision in industrial robotics. In addition, we examine integration challenges and propose strategies for implementing event camera technology in existing industrial infrastructure. This work contributes valuable insights into the future trajectory of adaptive and intuitive HRC systems, offering a roadmap for researchers and practitioners in the field of industrial automation.

Index Terms—Event camera, gesture recognition, human action recognition, human pose estimation, human–robot collaboration (HRC), neuromorphic vision.

I. INTRODUCTION

INDUSTRIAL robotics has undergone a transformative evolution: from isolated, caged robots focused solely on repetitive tasks, to systems capable of basic human–robot interaction (HRI), then advancing to collaborative robots that share workspaces. In this context, human–robot collaboration (HRC) has emerged as a pivotal area of research, particularly in industrial settings, where robots are expected to work alongside humans in a safe and efficient manner [1]. The integration of robots in such environments requires them to interpret human intent, actions, and behavior accurately to ensure seamless interaction and cooperation. Traditional methods for achieving HRC largely rely on vision-based systems, with red, green, blue (RGB) cameras being the most common tool for capturing visual data [2]. However, these cameras face significant limitations, especially in dynamic and complex environments.

RGB cameras are constrained by their dependence on ambient light, often struggling in low light or high-speed scenarios. They also fail to capture motion with high temporal precision, making it challenging to track rapid human gestures or actions [3]. Furthermore, RGB cameras capture frames at fixed intervals, leading to potential data loss in high-motion or fast-paced environments, which reduces the accuracy of human action recognition and interaction prediction.

In response to these challenges, event cameras have emerged as a promising alternative. Event cameras, or neuromorphic sensors, operate by capturing changes in the scene as individual events (pixels that change over time), offering several advantages over conventional cameras [4]. These devices can operate at high temporal resolution, detecting movements as small as microseconds, and they are highly sensitive to light changes, allowing them to function in both low-light and high-speed conditions. Furthermore, event cameras capture continuous streams of data, unlike traditional cameras that capture frames at fixed intervals, enabling real-time, high-precision tracking of dynamic actions.

In this systematic review paper, a comprehensive framework is proposed, as shown in Fig. 1. This highlights the potential of event cameras in enhancing HRC through the processing of various human states. The flow begins with the event camera capturing dynamic scenes, where each change is processed and analyzed to extract important features such as human pose, gestures, and facial expressions. These features provide essential

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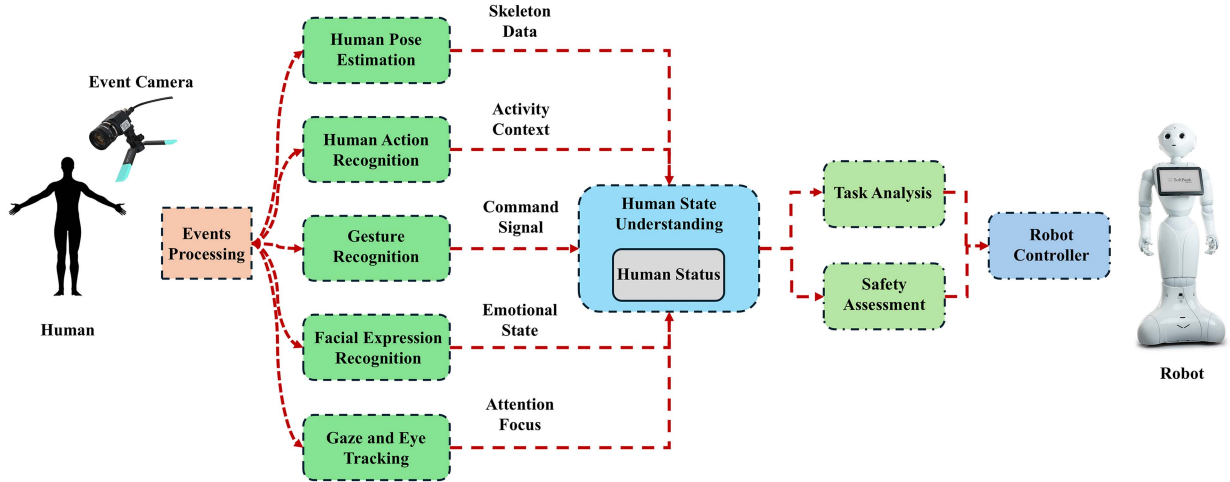


Fig. 1. Potential of event cameras in enhancing HRC through the processing of various human states, i.e., human pose estimation, human action recognition, gesture recognition, facial expression recognition (FER), and gaze and eye tracking.

cues for understanding the human's current status, including their pose, emotional state, activity context, and attention focus.

The human state understanding module synthesizes this information, resulting in a comprehensive representation of the human's status, which includes real-time information about their actions, gestures, and emotional state. This data is used to generate command signals, interpret the task at hand, and assess safety conditions. For instance, understanding a human's facial expression and emotional state could help determine if the human is in distress or if a safety intervention is required. Similarly, recognizing gestures and gaze direction can provide contextual information about the human's intentions, allowing the robot to respond appropriately. In the context of HRC, this understanding is crucial. It allows the robot to not only interpret and predict human actions but also adapt to the human's emotional and situational needs. The final step in the system involves the robot controller, which processes the command signals and ensures the robot performs the task with safety and efficiency in mind. By utilizing event cameras, this approach enables robots to achieve a higher level of situational awareness, safety, and task performance in environments that require close human-robot interaction. The combination of precise human state understanding and fast, responsive robotic control paves the way for more intuitive, adaptive, and efficient collaborative systems in real-world scenarios.

This review will explore the current research and applications in this domain, highlighting the role of event cameras in improving HRC and outlining the challenges and opportunities for future development.

A. Motivation

The motivation behind this work stems from the growing need for advanced HRC systems that can operate in dynamic and complex real-world environments. As robots are increasingly deployed in industrial, healthcare, and other collaborative settings, the demand for intuitive and responsive systems capable of understanding human actions and intentions in real time is critical. Traditional vision-based systems, especially RGB

cameras, struggle with the limitations of low-light conditions, high-speed movements, and the inability to capture fine-grained, continuous actions, hindering the effectiveness of human-robot interactions. Event cameras, with their high temporal resolution, low latency, and robustness to lighting variations, offer a transformative solution. By capturing asynchronous events at the pixel level, event cameras can track even the most rapid or subtle movements, enabling robots to interpret and respond to human gestures, poses, and expressions with unprecedented precision and speed. This work seeks to bridge the gap between the existing capabilities of event cameras and their untapped potential in enhancing HRC. By exploring the application of event cameras to human pose estimation, activity recognition, gesture recognition, FER, and gaze tracking, this research aims to provide a comprehensive overview of how these sensors can revolutionize robot perception, improving the safety, efficiency, and intuitiveness of human-robot interactions in collaborative settings.

II. METHODOLOGY FOR PAPER SELECTION

A. Paper Selection Process

The systematic literature review process followed the Preferred Reporting Items for Systematic Reviews and Meta-Analyses (PRISMA) guidelines, as shown in Fig. 2. The search was conducted across three major scientific databases: Web of Science ($n = 105$), Scopus ($n = 186$), and IEEE Xplore ($n = 156$), using six different search term combinations focused on neuromorphic vision and human behavior understanding applications.

The search terms were structured around the core concept of neuromorphic vision systems (including "neuromorphic," "event camera," and "dynamic vision sensor") combined with specific human behavior understanding applications: human pose estimation, human action/activity recognition, gesture recognition, eye tracking/gaze analysis, and emotions/expression recognition. The list of keywords used for the search is presented in Table I. This comprehensive search strategy yielded a total of 447 records across all databases.

The selection process consisted of three main phases:

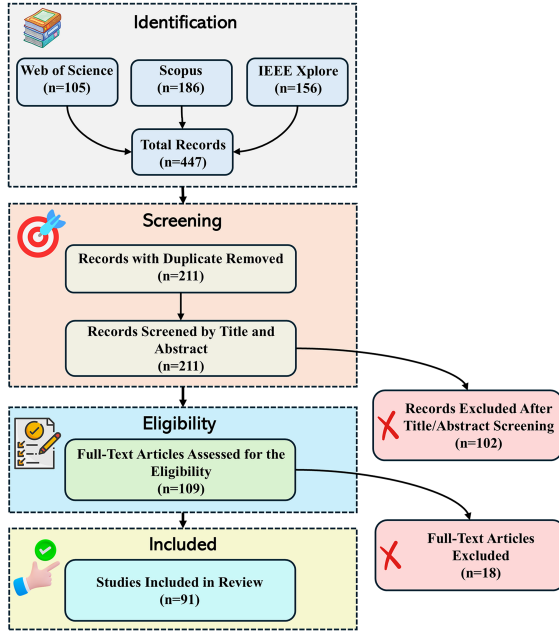


Fig. 2. PRISMA diagram of papers selection methodology adopted for this work.

TABLE I
SEARCH TERMS COMBINATIONS USED FOR REVIEW IN THIS WORK

	Search Terms
Term 1	("neuromorphic" OR "event camera" OR "dynamic Vision sensor")
Term 2	("human pose estimation") AND Term 1
Term 3	("human action recognition" OR "human activity recognition") AND Term 1
Term 4	("gesture recognition") AND Term 1
Term 5	("Eye Tracking" OR "Gaze") AND Term 1
Term 6	("Emotions Recognition" OR "Expression Recognition" OR "Human Emotions Recognition" OR "Facial Expression") AND Term 1

1) Screening Phase:

- Initially, duplicate records were identified and removed, reducing the total number of unique papers to 211.
- These papers were then screened based on their titles and abstracts, resulting in the exclusion of 102 papers that did not meet the review's scope or relevance criteria.

2) Eligibility Phase:

- The remaining 109 full-text articles were assessed for eligibility based on detailed inclusion and exclusion criteria.
- During this phase, 18 papers were excluded after full-text review for not meeting the specific eligibility requirements.

3) Inclusion Phase:

- The final selection resulted in 91 papers that fully met all inclusion criteria and were included in the systematic review.

This rigorous selection process ensured that only the most relevant and high-quality papers were included in the final review, providing a comprehensive yet focused analysis of the current state of neuromorphic vision applications in robotics with human in loop.

III. HUMAN POSE ESTIMATION

Event cameras have emerged as a promising alternative to traditional frame-based cameras for human motion capture and tracking applications. Unlike conventional cameras that capture frames at fixed intervals, event cameras feature independent pixels that asynchronously measure intensity changes. This design offers several key advantages, including high temporal resolution, high dynamic range, and low latency.

The foundation for event-based human tracking was established through methods for spatio-temporal tracking using Gaussian mixture models, which specifically addressed the challenges of tracking multiple persons with occlusions [5]. Building on this groundwork, a hybrid approach was developed that combined events with standard frames, implementing a generative event model within a maximum likelihood framework to achieve subpixel accuracy in feature tracking [6]. As the field progressed, researchers began developing more sophisticated systems. A notable contribution is EventCap, a system designed for 3-D motion capture primarily using event data, which demonstrated how events could enhance high-frequency motion estimation through batch optimization and convolutional neural network (CNN)-based pose detection [7]. The introduction of specialized datasets like DHP19 and Event-Human3.6M has further enabled rigorous evaluation of event-based methods [8]. Recent advances have pushed the boundaries of pure event-based approaches, such as a deep learning pipeline that aggregates events into tensor-like representations and uses multistage CNNs to predict orthogonal pose projections [9]. A novel pure event-based method was also demonstrated, where a single event camera is moved around a stationary subject to create 3-D reconstructions through voxel carving, eliminating the need for RGB input entirely [10]. Some researchers have explored hybrid systems that combine event and traditional cameras. For example, EvRGBHand integrates both modalities to achieve robust hand mesh reconstruction, using the event camera's high dynamic range to handle the RGB overexposure and RGB data to compensate for foreground scarcity in events [11]. In parallel, MoveENet was developed to achieve real-time performance at up to 100 Hz through an efficient neural network architecture and the exponentially reduced ordinal surface event representation [12].

Fig. 3 illustrates a comprehensive pipeline for real-time recognition of human pose, actions, and gestures using an event camera. The process begins with the input stage, where the event camera captures continuous motion in the form of an event stream rather than traditional image frames. This stream undergoes event preprocessing, where tasks such as denoising, polarity filtering, and frame time-windowing are applied. These steps help clean the data and organize it into a structured format like the event voxel grid, making it suitable for further analysis. The main objective of this stage is to convert raw asynchronous data into consistent, noise-free input while preserving critical temporal information.

Following preprocessing, the pipeline proceeds to the feature extraction stage, where spatio-temporal features are derived using three distinct approaches: deep learning, handcrafted methods, and neuromorphic computing. Deep learning models

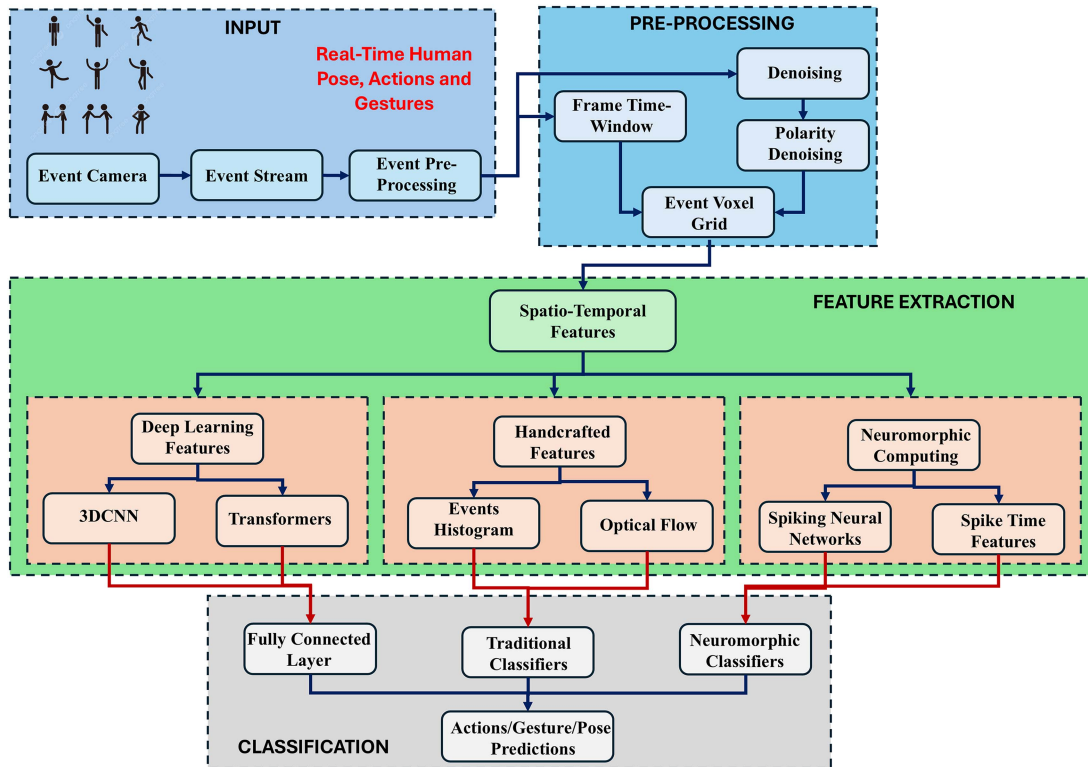


Fig. 3. Detailed flow of end-to-end pipeline for real-time human pose estimation, action, and gesture recognition using event cameras, incorporating preprocessing, spatio-temporal feature extraction through deep learning, handcrafted, and neuromorphic methods, followed by classification for accurate activity prediction.

TABLE II
SUMMARY OF DATASETS OF THE HUMAN POSE ESTIMATION USING THE EVENT CAMERA

Ref.	Type	Subjects	Movements	Resolution	Key Features
[13]	Event-based	Multiple	Various dynamic poses	Not specified	3-D human pose and shape estimation using event-based signals.
[8]	Event-based	17	33 movements, 10 reps	346 × 260 pixels	Real-time multiview 3-D human pose estimation using DVS.
[9]	Synthetic	Multiple	Various movements	Not specified	Simulated events from Human3.6m dataset for pose estimation.

such as 3-D convolutional neural networks (3DCNNs) and transformers automatically learn complex features from event data. In parallel, handcrafted features like event histograms and optical flow offer simpler, human-designed descriptors. Neuromorphic computing introduces brain-inspired techniques using spiking neural networks (SNNs) and spike-time features for event-based learning. These extracted features are then classified through appropriate models—fully connected layers for deep learning, traditional classifiers for handcrafted features, and neuromorphic classifiers for spike-based representations—ultimately enabling accurate predictions of human pose, actions, or gestures in real time.

A. Datasets

The datasets discussed in Table II provides valuable resources for event-based human pose and shape estimation, which are elaborated in detail further in this section. The EventHPE dataset [13] uses a novel two-stage deep learning approach, leveraging event camera data along with a first-frame grayscale image to estimate 3-D human poses and shapes. It includes various dynamic human poses and is the largest event-based 3-D human pose and shape dataset available. The DHP19 [8]

dataset offers a multiview setup with four synchronized Dynamic and Active-pixel Vision sensor (DAVIS) cameras, capturing 33 movements from 17 subjects in a controlled therapy environment, providing accurate 3-D pose estimations with a Vicon motion capture system. This dataset is focused on real time, low-latency applications for 3-D human pose estimation. The Event-Human3.6m dataset [9] is a synthetic dataset, generated by simulating event-based data from the standard Human3.6M dataset, which contains various complex human movements, making it an essential tool for benchmarking event-based pose estimation methods. These datasets offer diverse challenges and contribute significantly to advancing research in human pose estimation using event cameras.

B. Challenges

One of the most pressing challenges in event-based human pose estimation is the lack of diverse, large-scale datasets. While datasets such as DHP19 [8], Event-Human3.6M [9], and EventHPE [13] have contributed to benchmarking efforts, they remain relatively small and are often collected in controlled environments. This limitation affects the ability of models to generalize across real-world scenarios involving varying

TABLE III
COMPREHENSIVE OVERVIEW OF RECENT ADVANCEMENTS IN ACTIVITY RECOGNITION LEVERAGING NEUROMORPHIC/EVENT-BASED VISION SYSTEMS

Ref	Year	Method/Technique Used	Sensor/Dataset	Results Achieved
[16]	2017	Biologically inspired event-driven architecture	DVS128	Real-time classification at 1000 fps, 70 mW power consumption, better than key pose-based methods.
[17]	2017	Motion Maps + Motion Boundary Histogram fusion	DVS Gesture	Comparable to conventional video-based HAR approaches.
[18]	2019	Transfer learning on NVS with optical flow input	NVS Custom Dataset	Recognition within 8.8% of I3D w/ flow; faster frame generation, lower complexity.
[19]	2019	Event Memory Surface with CNN	Custom DVS Dataset	Achieved 94.3% accuracy on custom human activity dataset.
[20]	2019	Spatiotemporal filtering in spike-event domain	DVS Gesture	Boosted CNN performance using spatiotemporal filtering.
[21]	2020	Event-based feature extraction and classification	DVS Benchmark Dataset	Low-power human activity recognition.
[22]	2021	Timestamp image encoding with 2D CNN	DVS128	SOTA performance in gesture recognition, improved incomplete action handling.
[23]	2021	Handcrafted features with SVM and KNN	E-KTH, E-UCF11, E-HMDB51	Outperformed baseline methods on multiple event-based HAR datasets.
[24]	2021	SNN for motion feature extraction	DVS Gesture	Outperformed traditional SNNs on event-based datasets.
[25]	2022	Fine-tuning MViT for neuromorphic vision	DVS-Gesture, DVS128	Achieved 0.958 accuracy and F1-score.
[26]	2022	Aligned Compressed Event Tensor (ACE), BET network	DVS128 Action	98.88% accuracy, low power, outperformed other methods.
[27]	2022	Conv Leaky Integrate-and-Fire SNN	IITM-DVS, UCF11-DVS	Better accuracy than competing methods across datasets.
[28]	2022	Adaptive windowing + event density temporal localization	Fall Detection Custom Dataset	High-efficiency, privacy-preserving fall detection.
[29]	2022	Hybrid neural architecture (event + image)	Egocentric Event Dataset	Event frames outperform RGB images for egocentric action recognition.
[30]	2023	Spiking Transformers for wearable HAR	Wearable HAR Dataset	Comparable accuracy with major energy-delay product gains.
[31]	2023	Hybrid 3D-CNN + SNN	Indoor Surveillance Dataset	97.84% accuracy for fall detection.
[32]	2022	Event-based 3D ResNet-like model	N-EPIC Kitchens Dataset	Up to 4% better accuracy than RGB-only models.
[33]	2023	TTPOINT: Lightweight point-cloud-based network	Custom Event Dataset	SOTA accuracy with only 1.5% of computation compared to conventional methods.
[34]	2023	Video Transformer + Event Contrastive Loss	DVS Gesture, N-EPIC Kitchens	SOTA performance with reduced computation time.
[35]	2024	Spike-HAR	Spike-HAR++ Dataset	Top performance across 4 datasets; ultra-low power.
[36]	2024	HyperMV: Multi-view action recognition framework	Multi-view DVS Datasets	Outperforms baselines in cross-view and cross-subject tests.
[37]	2024	Bayesian Neural Networks	EventSleep Dataset	Robust HAR under low-light using event cameras.
[38]	2024	SpikMamba SNN Framework	Multiple HAR Event Datasets	SOTA recognition results surpassing previous methods.
[39]	2024	EventCrab: Synergy-aware dual network	SeAct, HARDVS	Improved accuracy combining frame- and point-based learning.
[40]	2024	ESTF: Spatial-temporal Transformer	HARDVS	Strong results with detailed baseline comparisons.

lighting conditions, occlusions, and diverse human movements. Another major issue is spatial and temporal misalignment between event-based data and ground truth labels [14]. Unlike RGB cameras, which capture full-frame images at fixed intervals, event cameras operate asynchronously, making it difficult to ensure that labeled keypoints in the dataset correspond accurately to the motion captured in event streams. In addition, computational efficiency remains a significant barrier to real-time deployment. Deep learning-based pose estimation models, especially those relying on CNNs and transformers, require high computational power, which makes them unsuitable for edge devices and embedded systems commonly used in robotics and wearable applications [7]. While SNNs and event-driven architectures have been proposed, they are still in the early stages of development and require further optimization to balance accuracy and efficiency [15].

IV. ACTIVITY RECOGNITION

The primary challenge in working with event cameras lies in handling their unique data output, which consists of asynchronous, event-based information rather than the traditional frame-based data. This type of data captures changes in the scene at the level of individual pixels, making it more efficient for dynamic environments but also more complex to process and integrate into conventional computer vision systems. Several approaches have been developed to address this, particularly in

event aggregation techniques that transform asynchronous data into synchronized visual frames. To address these differences, several approaches, such as those in [20] and [41], have concentrated on developing event aggregation techniques. While earlier methods relied on handcrafted features [17], [42], deep learning-based approaches have gained significant traction in recent years. For instance, binary representations were created by evaluating the presence or absence of events at each pixel within a fixed time interval Δt , followed by a binary-to-decimal conversion process for frame generation [41].

A significant breakthrough in the field came with the introduction of biologically inspired visual architectures for analyzing articulated human movements. The model uses a deep CNN that demonstrates real-time action recognition capabilities while maintaining remarkable energy efficiency, operating at approximately 1000 frames/s while consuming only 70 mW of power [16]. This approach has been complemented by methods combining dynamic vision sensor (DVS) data with motion maps and motion boundary histograms (MBHs), achieving performance comparable to traditional video-based methods in human action recognition (HAR) tasks [17]. The integration of a neuromorphic vision sensor (NVS) with advanced deep learning architectures has significantly enhanced action recognition performance. Event memory surfaces have been adapted to make sparse event data compatible with CNNs, resulting in high accuracy on newly introduced human activity datasets [19]. Another approach addresses the challenge of noise and sparse

information in event-based data by introducing spatiotemporal filtering, which enhances CNN performance and improves the recognition of complex actions [20]. SNNs have also been explored as a solution for processing motion features in event-based cameras, offering a hierarchical structure that captures motion information from local to global levels [24]. Despite these advancements, significant challenges remain in encoding and processing spatiotemporal data effectively for complex recognition tasks, particularly in real-world scenarios involving diverse and unpredictable human actions [21], [22].

Several studies have explored novel representations and architectures for processing event data, such as the aligned compressed event tensor (ACE) and the branched event network (BET). The ACE-BET framework has been reported to outperform state-of-the-art methods by significant margins, achieving 98.88% accuracy for action recognition using the DVS128 dataset [26]. SNNs have also been utilized to address the noise and sparsity challenges inherent in event-based data [27]. In addition, for privacy-preserving applications, event cameras have been proposed for fall detection, where the combination of temporal localization frameworks with spatiotemporal attention mechanisms has led to significant improvements in performance [28], [31]. Event cameras have also shown promise in egocentric action recognition, particularly in wearable applications. Event-based visual data has been compared with traditional images, revealing that event frames provide discriminative information that enhances action recognition performance [29]. Moreover, SNNs have been explored for energy-efficient human activity recognition (HAR) tasks [30]. Another noteworthy contribution is the introduction of the N-EPIC-Kitchens dataset [32]. One of the most significant innovations is the development of the TTPOINT framework [33]. In addition, the Spike-HAR and Spike-HAR++ models introduce spiking transformers for event-based HAR [35]. Moreover, the EventCrab framework integrates the strengths of frame-specific and point-specific networks [39]. The field has also seen significant progress in the development of datasets and methodologies. Furthermore, studies focusing on multiview event-based action recognition have demonstrated effectiveness through the HyperMV framework [36]. In addition, datasets like EventSleep have showcased the potential of event cameras in medical applications [37]. The SpikePoint architecture represents another significant advancement [43]. Another key advancement is the application of event cameras to more fine-grained tasks such as facial microexpression recognition, with a novel spatio-temporal vision transformer (ViT) model trained on the FACEMORPHIC dataset [44].

The field continues to evolve, with ongoing research focusing on optimizing algorithms to handle the sparse, asynchronous nature of event data and extending these systems to larger scale, real-world scenarios. The summary of the event camera works on HAR is also provided in Table III, which elaborates the methods/techniques used by the previous works with the sensor/dataset used and the results achieved.

A. Datasets

Table IV summarizes several event-based activity recognition datasets that are crucial for advancing research in HAR using

TABLE IV
SUMMARY OF PUBLICLY AVAILABLE EVENT-BASED ACTIVITY RECOGNITION DATASETS

Ref.	Sensor	Samples	Resolution	Class
[45]	DAVIS346	450	346 × 260	10
[24]	DAVIS346	1440	346 × 260	12
[46]	DAVIS346	300	346 × 260	10
[40]	DAVIS346	100,000	346 × 260	300
[47]	CeleX-V	10,500	640 × 480	114
[48]	CeleX-V	27,102	640 × 480	50
[49]	DVXplorer Lite	22,046	320 × 240	200
[50]	CeleX-V	124,625	1280 × 800	150

neuromorphic vision systems. PAF dataset was recorded using the DAVIS346 sensor, features 450 samples at a resolution of 346 × 260 across ten activity classes [45]. Daily Action, also recorded with the DAVIS346 sensor, offers 1440 samples across 12 activity classes, further contributing to diverse action recognition tasks [24]. The Bully10K dataset, with 300 samples from the DAVIS346, focuses on ten activity classes [46]. HARDVS offers the largest sample size at 100 000, recorded with the DAVIS346 sensor, and contains 300 classes, making it a comprehensive resource for complex activity recognition [40]. PokerEvent, captured with the CeleX-V sensor, provides 10 500 samples across 114 classes, ideal for gesture-based interaction recognition [47]. The THUE-^{ACT}50 [48] dataset, recorded with CeleX-V, includes 27 102 samples and 50 activity classes, catering to various activity recognition applications. Dailydvs-200 captured using the DVXplorer Lite sensor, provides 22 046 samples across 200 classes, offering detailed recognition tasks for dynamic activities [49]. Finally, CeleX-HAR [50], the largest dataset with 124 625 samples, recorded with CeleX-V, encompasses 150 classes, enabling advanced HAR model training. These datasets showcase a variety of sensors, sample sizes, resolutions, and activity classes, providing valuable resources for the development and evaluation of event-based activity recognition systems.

B. Challenges

Event-based activity recognition presents unique challenges due to the sparse, asynchronous nature of event data. Traditional activity recognition models rely on dense spatio-temporal information from RGB videos [51], whereas event cameras provide nonuniform motion-based data, making direct adaptation of conventional methods difficult. One of the key challenges is choosing an optimal event representation technique. While some approaches convert event data into frame-like representations using voxel grids, motion tensors, or time surfaces, these methods often lead to information loss or increased computational complexity [52]. On the other hand, purely event-based processing methods, such as SNNs, require specialized hardware and are not yet widely adopted [53]. In addition, event cameras are sensitive to background motion and noise, which can affect activity recognition accuracy in dynamic environments such as industrial workplaces or crowded public spaces [4]. Current models struggle with distinguishing between meaningful human actions and irrelevant background events, leading to increased false positives. Another challenge is real-time processing for wearable and edge applications. Many potential use cases, such as health monitoring and assistive robotics, require low-power, real-time

inference—a goal that remains difficult due to the high computational demands of deep learning-based event processing models.

V. GESTURE RECOGNITION

Event-based cameras have demonstrated significant potential in gesture recognition applications, offering advantages over traditional frame-based methods. Foundational approaches were established using DVS technology, such as a biologically inspired method employing leaky integrate-and-fire (LIF) neurons for tracking finger movements and detecting swipe gestures with minimal latency [54]. This work was extended with a system that utilized stereo DVS sensors and neuromorphic postprocessing to achieve robust gesture recognition while addressing challenges such as gesture spotting and background noise [55].

A key advancement in event-based processing has been the development of algorithms that harness the spatio-temporal nature of event streams. For instance, event streams have been treated as space-time point clouds, with the PointNet architecture used to achieve strong gesture recognition performance [58]. This was further improved through the use of dynamic graph CNN (DGCNN), which analyzes the 3-D geometry of event clouds and has shown high accuracy on datasets such as IBM DVS128 and DHP19 [61]. Deep learning has also played a critical role, including a CNN-based demosaicing method that enhances far-distance gesture recognition by leveraging spatiotemporal correlations [56], and a FlowNet-based approach that improves motion representation while demonstrating the efficiency and data-reduction benefits of event cameras over conventional frame-based systems [59].

The integration of neuromorphic hardware has facilitated low-power, real-time processing for event-based systems. A notable example is an end-to-end system developed using the TrueNorth neurosynaptic processor, which processes DVS event streams in real time with impressive performance and minimal power consumption [57]. In mobile and embedded environments, real-time gesture recognition has been achieved using an Android-based framework that includes background suppression techniques [60]. Innovative methods for event data processing have also emerged, such as an event aggregation strategy based on temporal binary representation [41] and a wearable glove enhanced with high-frequency LED markers to improve tracking accuracy [62]. Research into lightweight and efficient models has led to systems based on liquid state machines [63] and applications of DVS in prosthetic vision [64]. The role of domain adaptation and transfer learning has gained prominence in making event-based systems more versatile. An unsupervised spatio-temporal domain adaptation framework using deep SNNs has shown substantial improvements in cross-domain performance [65]. In addition, integrating neuromorphic systems with blockchain has been proposed as a decentralized model for intelligent neuromorphic research [66]. Industrial and real-time applications are also advancing, as seen in the brain-inspired spiking gating flow system for enhanced action recognition [67] and optimized data preprocessing techniques for embedded systems [68]. Furthermore, few-shot learning is emerging as a vital area, with an energy-efficient framework

utilizing surrogate gradient-based SNNs introduced to address data scarcity challenges [69].

Recent developments have increasingly focused on specialized applications of event-based vision, such as sign language recognition. An event-based sign language dataset was introduced, achieving 77% recognition accuracy using SNNs [70]. Hardware efficiency has also been explored through quantization and pruning techniques applied to gesture recognition models [71], along with an ultra-low power implementation using the Kraken PULP system-on-chip [72]. Efforts to optimize data processing have led to a cascaded, timestamp-free compression method that reduces data redundancy without sacrificing recognition accuracy [73], while dynamic human-object interactions have been investigated through the release of a new dataset for hand-object action recognition [74].

Event cameras have also found promising roles in educational and assistive technologies. For instance, their potential in supporting visually impaired students has been explored in an educational context [75]. The field of sign language recognition has been further refined through optimized SNN architectures that improve both training speed and memory efficiency [76]. Robotic systems have benefited from high-resolution event cameras to improve guidance and interaction capabilities [77]. Advanced gesture recognition in complex and dynamic environments has been addressed using multidimensional projection and spatiotemporal event frames [78]. In addition, significant progress has been made in energy-efficient algorithm-hardware co-design, achieving high accuracy while maintaining minimal energy usage [79].

Table V shows a summary of recent work done using the event camera for gesture recognition which elaborates the methods/techniques used by the previous works with the sensor/dataset used and the results achieved. These developments collectively demonstrate the rapid evolution and diverse applications of event-based cameras in gesture recognition, from fundamental algorithmic advances to practical implementations in various domains. The field continues to progress toward more efficient, robust, and practical solutions for real-world applications.

A. Dataset

The event-based gesture recognition datasets represent significant contributions to the field of neuromorphic vision and gesture recognition, presented in Table VI. The EB-HandGesture dataset, collected using the SilkyCam Gen3, features high-resolution data (640×480) with six distinct gestures, offering valuable material for hand gesture recognition in dynamic real-world environments [77]. The DVS128 Gesture dataset, captured with the DAVIS128 sensor, comprises ten gestures designed to evaluate recognition performance in controlled settings, with each instance lasting 6 s [57]. The Nav-DVS dataset, recorded using the ATIS sensor, emphasizes six gestures in real world, moving camera scenarios [60]. The ASL-DVS dataset, obtained with the DAVIS240 sensor, contains over 100 000 samples of American Sign Language gestures, providing a rich resource for sign language recognition [80]. Collectively, these

TABLE V
COMPREHENSIVE OVERVIEW OF RECENT ADVANCEMENTS IN GESTURE RECOGNITION LEVERAGING NEUROMORPHIC/EVENT-BASED VISION SYSTEMS

Ref.	Year	Method/Technique Used	Sensor/Dataset Used	Results Achieved
[54]	2012	LIF neurons for fingertip/swipe tracking	DVS (unspecified)	Real-time UI tracking with millisecond intervals and negligible latency
[55]	2014	SNN + HMM on stereo event data	Stereo DVS	>90% recognition accuracy under various conditions
[56]	2016	CNN-based demosaicing for gesture frames	DVS Camera	Enhanced distant gesture recognition performance
[57]	2017	CNN on IBM TrueNorth processor	DVS128/TrueNorth	96.5% accuracy with only 200mW power consumption
[58]	2019	PointNet on event cloud data	IBM DVS128	97.08% accuracy
[59]	2019	FlowNet2.0 + CNN for motion encoding	Event Camera (unspecified)	94.0% accuracy @ 30 fps
[60]	2020	Time-surfaces on Android platform	Smartphone Event Sensor	Real-time operation comparable to SOTA
[61]	2020	Dynamic Graph CNN (DGCNN)	IBM DVS128, DHP19	98.56% (IBM DVS128), 95.94% (DHP19)
[41]	2021	Temporal Binary Representation (TBR)	DVS128 Gesture	Achieved SOTA results
[62]	2021	Wearable LED + particle filter tracking	Custom wearable DVS setup	Robust against illumination changes
[63]	2022	Liquid State Machine + heuristic search	DVS Gesture Dataset	98.42% accuracy; 90% fewer parameters than Inception3D
[64]	2022	DVS-based prosthetic vision interface	DVS with AI prosthetics	90% accuracy in gesture/object recognition
[65]	2022	Domain-adaptive SNN with gradient reversal	Multiple Gesture Datasets	Up to 10.39% improvement across domains
[66]	2022	Blockchain-based SNN governance	Event-driven Neuromorphic Systems	Secured collaborative model behavior tracking
[67]	2022	Spiking Gating Flow (SGF)	Event Camera (unspecified)	87.5% accuracy with only one training epoch
[68]	2022	Event accumulation + overlapping frames	DVS Camera	94.6% accuracy, low memory and latency
[69]	2022	Surrogate gradient-based few-shot SNN	Gesture dataset (unspecified)	92.82% accuracy with reduced spike emissions
[70]	2023	SNN for sign language recognition	Sign Language Event Dataset	77% classification accuracy
[71]	2023	SNN quantization and pruning	Custom Gesture Dataset	Maintained accuracy, ternary weights; pruning increased energy consumption
[72]	2023	Ternary TCN on Kraken PULP SoC	Kraken PULP Chip + DVS	97.7% accuracy @ 4.7mW
[73]	2024	CTFC compression method	DVS Gesture Dataset	94.62% accuracy, 68% data size reduction
[74]	2024	Residual SNN with attention for H-O gestures	HO Event Dataset	Competitive with standard baselines
[75]	2024	Event-based prosthetic vision for education	Event Cameras for Visually Impaired	Enhanced gesture perception in classroom settings
[76]	2024	STBP, SLAYER, DECOLLE optimized SNNs	Event Sign Language Dataset	Improved training speed, memory usage
[77]	2024	ConvRNN network	EB-HandGesture Dataset	95.7% accuracy in varied lighting
[78]	2024	Spatiotemporal Event Frame (STEF) projection	Custom Event Dataset	96.67% accuracy
[79]	2024	CNN with median filtering on edge device	Custom Edge Platform	92.4% accuracy; 593.4nJ per inference

TABLE VI
SUMMARY OF PUBLICLY AVAILABLE EVENT-BASED GESTURE RECOGNITION DATASETS

Ref.	Sensor	Resolution	Duration	Samples	Gestures
[77]	Gen3	640×480	0.5s	1.5k	6
[57]	DAVIS128	128×128	6s	1.3k	10
[60]	ATIS	302×245	-	1.3k	6
[80]	DAVIS240	240×180	0.1s	100k	Various

datasets demonstrate the versatility of event-based cameras in gesture recognition, highlighting their capacity to capture dynamic movements with high temporal resolution and low data redundancy—qualities well-suited for applications in human–robot interaction, assistive technologies, and real-time systems.

B. Challenges

Event-based gesture recognition has made significant progress, but domain adaptation and generalization remain key challenges. Gesture models trained on controlled datasets often fail to perform well in unconstrained environments, where variations in lighting, camera angle, and user-specific differences (e.g., hand size, motion speed) affect recognition accuracy. This issue is further compounded by the lack of standardized, large-scale event-based gesture datasets. While datasets like IBM DVS128 [57], EB-HandGesture [77], and ASL-DVS [80] exist, they are significantly smaller than their RGB counterparts [81], [82] and contain a limited variety of gestures. In addition, most event-based gesture recognition systems require precise sensor positioning to achieve high accuracy, making them less adaptable for real-world applications where hand gestures may not always be perfectly centered in the camera’s field of

view. Another major challenge is computational efficiency and energy consumption, particularly for real-time gesture recognition in mobile or embedded devices. While event cameras inherently provide low-latency, high-speed motion data, processing this data with deep learning models is still computationally expensive [77]. Exploring lightweight architectures, SNNs, and neuromorphic hardware could improve efficiency, but these areas require further research.

VI. FACIAL EXPRESSION RECOGNITION

Facial expression recognition (FER) has emerged as a critical component in applications such as affective computing, robotics, and surveillance, enabling machines to interpret human emotions accurately. While deep artificial neural networks (ANNs) have revolutionized FER with high accuracy and robustness across various lighting conditions and head poses, they face significant challenges related to computational efficiency and energy consumption. As FER applications expand to resource-constrained environments such as edge devices and embedded systems, there is an increasing demand for energy efficient and scalable solutions. Recent research has introduced various approaches to address these challenges. An 18-layer ResNet model incorporating a novel Noisy Softplus (NSP) activation function was proposed to enhance antinoise capabilities and enable energy-efficient conversion of deep ANNs into SNNs, achieving high recognition accuracy on datasets like KDEF and GENKI-4K while reducing power consumption on neuromorphic hardware [83]. In a contrasting approach, a lightweight and practical FER solution was developed using histogram of oriented gradients (HOGs) and support vector machines

(SVMs), offering improved efficiency and compact storage while delivering performance comparable to deep learning methods [85].

Neuromorphic computing has emerged as a promising alternative for overcoming the high energy demands of traditional deep learning-based FER systems by leveraging biologically inspired SNNs that mimic the brain's energy-efficient processing. A comprehensive comparison between neuromorphic processors like Intel Loihi and edge AI devices such as Raspberry Pi-4, Intel Neural Compute Stick, and Jetson Nano revealed that neuromorphic systems can reduce power dissipation by up to two orders of magnitude without compromising accuracy [91]. Further extending this line of work, a SNN framework was applied to a DVS-based emotion recognition task, showing that it could maintain strong recognition performance while consuming only a quarter of the energy required by conventional ANN models [87].

Memristive networks have also been explored as an environmentally friendly and energy-efficient alternative for human emotion classification. An energy-efficient memristive sequencer network (EMSN) was introduced, utilizing 2-D materials in its hardware design to support sustainable consumer health monitoring applications [89]. Experimental results demonstrated that EMSN offers notable advantages in both computational efficiency and classification accuracy, making it a promising solution for scenarios with limited energy resources. Event cameras have gained significant attention as an alternative to traditional RGB cameras in FER applications due to their high temporal resolution, low latency, and energy efficiency. Unlike conventional cameras, event cameras respond to local changes in intensity, enabling them to capture microexpressions with greater precision and under challenging lighting conditions. In the context of Industry 4.0, Becattini et al. [86] demonstrated how event cameras can enhance human-machine interaction by providing real-time analysis of facial expressions while preserving privacy. Their research showed that by leveraging the event camera's ability to operate in dynamic environments, FER systems can achieve improved effectiveness and robustness compared to traditional RGB-based approaches.

Recent advances in event-based FER have highlighted the unique strengths of neuromorphic vision systems. A notable contribution is the introduction of the NEFER dataset, which includes paired RGB and event-based videos annotated with facial landmarks and emotion labels [88]. This dataset has enabled detailed comparisons between conventional and event-based approaches, with findings showing that event-based methods are better suited for capturing subtle microexpressions, making them ideal for analyzing rapid and hard-to-detect facial cues. Building on this progress, an event-based FER model called "SpikingFER" was developed to leverage the efficiency of SNN [92]. This model achieves energy consumption significantly lower than that of traditional ANN-based systems while maintaining comparable recognition accuracy. Together, these works demonstrate the strong potential of integrating event cameras with neuromorphic computing to develop highly efficient, low-power FER systems particularly valuable for deployment in mobile or edge computing environments.

Fig. 4 presents a complete system for real-time face analysis using an event camera, highlighting how facial cues can be captured and interpreted efficiently. The process begins with the input from an event camera, which records changes in the visual scene, particularly focusing on facial movements. The data pass through a preprocessing stage, where noise is removed, faces are detected, and important facial landmarks such as eyes and other expression-related areas are identified. These regions of interest (ROIs) are critical for narrowing down where to analyze detailed facial features, ensuring the system focuses on the most informative parts of the face.

Once the facial regions are identified, the pipeline splits into two main analysis paths: eye analysis and expression analysis. The eye analysis block tracks eye movements by detecting the pupil and corneal reflection, allowing the system to estimate where a person is looking (gaze estimation). On the other side, expression analysis handles both macro and microfacial expressions. It detects facial action units (like raised eyebrows or tightened lips) and fast, subtle microexpressions that reveal emotions. These features are then processed using machine learning techniques to interpret a person's attention level, emotional state, and facial expression. This system offers a nonintrusive and real-time way to understand human behavior and emotional responses, making it useful for applications in human-computer interaction, healthcare, and social robotics.

Table VII shows the summary of recent works done for FER using the event camera, which elaborates the methods/techniques used by the previous works with the sensor/dataset used and the results achieved. These developments collectively demonstrate the evolution of FER systems toward more efficient and practical solutions, particularly through the integration of neuromorphic computing, event-based vision, and novel hardware architectures. The field continues to advance toward achieving a balance between recognition accuracy and energy efficiency, making FER systems more viable for real-world applications in resource-constrained environments.

A. Datasets

Table VIII summarizes key datasets used for emotion and FER with event cameras, showcasing their characteristics such as the number of videos, users, resolution, sensor type, and available annotations. These datasets enable advancements in neuromorphic vision, where event cameras offer high-speed recognition, low latency, and robustness to lighting variations. Each dataset serves a unique purpose, from face tracking and alignment to detailed emotion analysis.

The Savran and Lenz datasets, both recorded using ATIS event cameras, focus on face alignment and tracking tasks [99], [100]. The Savran dataset includes recordings of head rotations with varying motion intensities, making it suitable for coarse face alignment in dynamic environments [99]. In contrast, the Lenz dataset introduces a novel approach by using eye blinks as natural facial motion cues to detect and track faces in an event-driven manner [100]. The dataset by Becattini, captured with a VGA event camera, is specifically tailored for microexpression recognition, enabling the detection of subtle and rapid

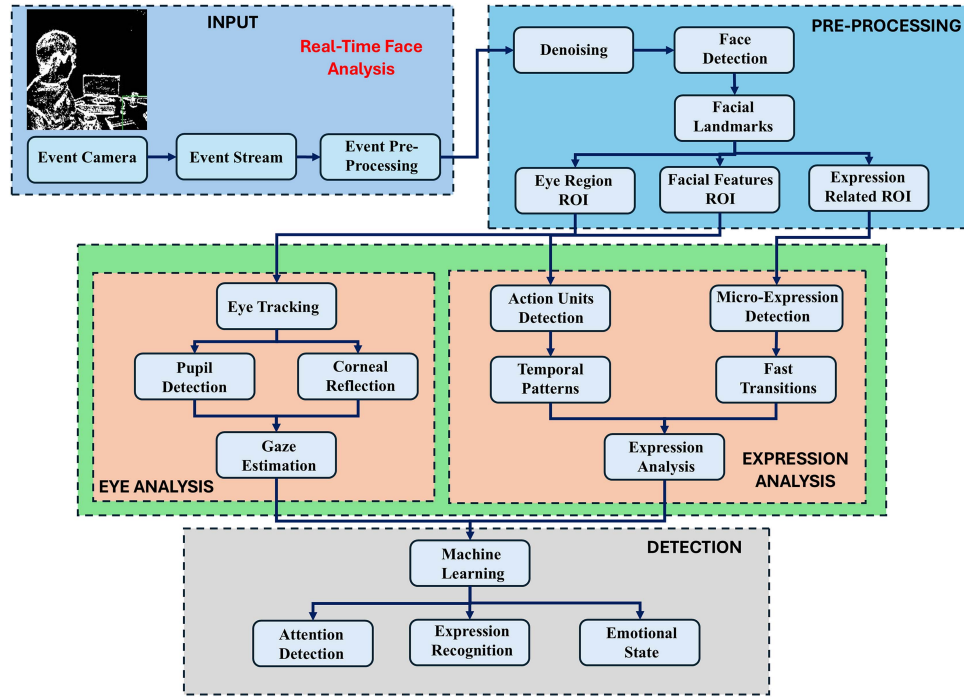


Fig. 4. Detailed flow for real-time face analysis using event cameras, incorporating preprocessing for facial region extraction, followed by eye tracking and expression analysis modules, which enable gaze estimation, action unit detection, and microexpression recognition.

TABLE VII
COMPREHENSIVE OVERVIEW OF RECENT ADVANCEMENTS IN FER LEVERAGING NEUROMORPHIC/EVENT-BASED VISION SYSTEMS

Ref.	Year	Method/Technique Used	Sensor/Dataset Used	Results Achieved
[83]	2018	ResNet with Noisy Softplus (NSP) activation	Simulated DVS (FER dataset)	Improved anti-noise robustness, reduced power consumption
[84]	2021	BSN with Viola-Jones and LBP features	Event-based simulated emotion dataset	69.5% accuracy across six emotions
[85]	2021	HOG + SVM with Genetic Algorithm optimization	DVS Camera (custom setup)	Competitive with DL, less storage and power
[86]	2022	Event-based architecture for micro-expression recognition	VGA Event Camera	Robustness under poor lighting; accurate subtle cue detection
[87]	2023	Spiking NN with DVS simulation for FER	Simulated DVS data	25% of ANN energy; performance retained
[88]	2024	NEFER Dataset contribution	NEFER Dataset (DVS, high-res)	Doubled accuracy vs. RGB data
[89]	2023	Memristive Sequencer Network (EMSN)	Simulated Event Data (Facial expression)	Higher computational efficiency and accuracy
[90]	2023	Event-based facial/gesture expression analysis pipeline	DVS Camera (unspecified)	Better deformable object detection than traditional vision
[91]	2023	Comparison: Neuromorphic vs. Edge AI accelerators	Neuromorphic Processor (DVS input)	2× lower power, similar accuracy
[92]	2023	SpikingFER – SNN-based FER framework	Event-based facial expression dataset	65.39× energy reduction vs. ANN
[93]	2023	Spiking modality for multimodal gesture-emotion recognition	Multimodal Gesture Emotion Dataset (event + audio)	Enhanced emotion classification performance
[94]	2024	Micro-expression system using event-driven encoding	Custom event-based micro-expression dataset	High temporal fidelity for rapid expressions
[95]	2024	ESTME: Spatio-temporal motion enhancement network	Microexpression event camera data	High accuracy in subtle micro-expression detection
[96]	2023	Transfer learning across synthetic and real neuromorphic data	Simulated & real DVS for valence-arousal	Transferable models for valence/arousal estimation
[97]	2024	Temporal polar pixel-wise interaction module	Event-based micro-expression dataset	Boost in performance on micro-expression benchmarks
[98]	2025	HI-Net: Hierarchical Event-RGB Interaction Network	Hybrid Event+RGB Dataset	Improved single-eye expression detection in poor light

TABLE VIII
SUMMARY OF PUBLICLY AVAILABLE DATASETS USED FOR EMOTION OR EXPRESSION RECOGNITION WITH EVENT CAMERAS

Ref.	Videos	Users	Resolution	Sensor Type	BBox	Landmarks	Emotions
[99]	108	30	304 × 204	ATIS Event Camera	No	No	No
[100]	48	10	640 × 480	ATIS Event Camera	No	No	No
[86]	455	25	640 × 480	VGA Event Camera	No	No	No
[88]	609	29	1280 × 720	High-Resolution Event Camera	Yes	Yes	Yes
[101]	2405	111	DAVIS346	DAVIS346 Event Camera	Yes	Yes	Yes

facial expressions that traditional RGB cameras often fail to capture [86]. This makes it particularly valuable for applications in interactive and industrial settings, where recognizing nuanced emotional cues can significantly enhance HRC.

The NEFER dataset, acquired using a high-resolution event camera, stands out as one of the most comprehensive resources for FER, providing face bounding boxes, facial landmarks, and emotion labels to support deep learning in neuromorphic emotion recognition [88]. Similarly, the dataset by Zhang, recorded with the DAVIS346 event camera, is designed for single-eye emotion recognition and is especially relevant for compact, low-power wearable systems [101]. It captures a diverse range of emotional expressions under varying lighting conditions, making it ideal for real-world affective computing and human-machine interaction applications. Collectively, these datasets underscore the growing role of event cameras in advancing facial analysis, emotion detection, and neuromorphic computing.

B. Challenges

FER using event cameras introduces unique challenges, particularly in capturing microexpressions with high temporal precision [86]. Event cameras are excellent at detecting subtle changes in intensity, but facial deformations during microexpressions generate low-intensity events, making it difficult for models to distinguish between fine-grained expressions such as subtle smirks or eye twitches. In addition, the lack of large, diverse event-based facial datasets hinders the development of robust models. While datasets such as NEFER [88] and Event-FER [86] offer valuable benchmarks, they do not cover a broad range of human expressions, ethnicities, age groups, and environmental conditions, limiting the generalizability of trained models. Another key challenge is multimodal fusion with RGB and depth data. While hybrid models that integrate RGB and event-based data (e.g., EvRGBHand) show promise, effective fusion remains an open problem [11]. Synchronization issues between event and RGB modalities, differences in spatial resolution, and increased computational overhead make it difficult to develop efficient multimodal systems [102]. Moreover, privacy and ethical concerns arise when using event cameras for emotion analysis in real-world scenarios, particularly in sensitive applications like healthcare and surveillance. Addressing these challenges will require better dataset collection methodologies, improved feature extraction techniques, and real-time privacy-preserving algorithms.

VII. GAZE AND EYE TRACKING

Gaze tracking has seen significant advancements with the introduction of event cameras, offering high temporal resolution, low latency, and power-efficient solutions for real-time applications. Various studies have explored different methodologies to optimize gaze tracking for mobile and embedded systems. A real-time gaze tracking algorithm demonstrated the ability to operate at 30 Hz on a mobile processor while requiring only 30K parameters. By employing an Auto ROI mode that emulates an event camera, the system significantly reduces computational

overhead while achieving gaze accuracies of $0.1^\circ - 0.5^\circ$ [103]. Another approach utilizes a hybrid frame-event-based system to push update rates beyond 10,000 Hz, enabling ultra-low-latency gaze tracking with accuracy comparable to high-end commercial trackers [104]. Similarly, a CNN-based pipeline was introduced, which integrates event-based eye tracking with a pupil tracking mechanism, reducing CNN inference by 96% while maintaining an error of only 3.68 pixels [105].

Neuromorphic processing and SNNs have also been explored to enhance efficiency in event-based gaze estimation. A sparse change-based convolutional long short-term memory (CB-ConvLSTM) model was proposed to process event streams efficiently, reducing arithmetic operations by $4.7\times$ while maintaining high tracking accuracy [106]. A neuromorphic processor was used to run an SNN-based eye-tracking model called “Retina,” which achieved a pupil tracking error of 3.24 pixels while consuming minimal power [107]. A wake-up-based power-efficient gaze estimation system further improved energy efficiency by adapting processing levels based on the eye’s state, demonstrating potential for low-power applications in wearable devices [108].

The availability of large-scale event-based datasets has played a crucial role in benchmarking gaze tracking algorithms. The EV-Eye dataset, which comprises 1.5 million near-eye grayscale images and 2.7 billion event samples, provides a robust resource for evaluating multimodal eye-tracking methods [109]. In addition, RGBE-Gaze, a dataset collected from 66 participants, synchronizes RGB and event camera data to enable high-frequency remote gaze tracking [110]. These datasets have enabled extensive evaluations and comparisons across different gaze-tracking techniques, contributing to advancements in accuracy and robustness.

Synthetic data generation has also emerged as a viable alternative to address the limited availability of real event camera datasets. Synthetic event data have been shown to closely replicate real event data for gaze tracking, particularly in predicting saccadic eye movements. A dual ResUNet architecture was proposed to process synthetically encoded events, achieving 99.20% accuracy within a 100-pixel radius [111]. In addition, an RGB-to-event domain translation technique using the StyleFlow algorithm was developed to generate event-like images from RGB inputs, leading to improved pupil localization accuracy in cross-modal learning applications [112].

The integration of event cameras into healthcare and augmented reality/virtual reality (AR/VR) applications has also gained momentum. A query-driven dynamic vision sensor (qDVS) system was proposed for eye tracking in neurological studies, achieving over 93% accuracy in horizontal and vertical gaze movement estimation under various illumination conditions [113]. Swift-Eye, a high-frequency gaze tracking framework, was developed to support applications such as mental health diagnostics and foveated rendering in virtual reality, demonstrating strong robustness even under significant pupil occlusion [114]. In addition, a near-eye DVS-based gaze processor was introduced for AR/VR environments, achieving 0.91° accuracy at a dynamic frame rate of 1250 Hz while maintaining low power consumption [115].

TABLE IX
COMPREHENSIVE OVERVIEW OF RECENT ADVANCEMENTS IN GAZE AND EYE TRACKING LEVERAGING NEUROMORPHIC/EVENT-BASED VISION SYSTEMS

Ref.	Year	Method/Technique Used	Sensor/Dataset Used	Results Achieved
[103]	2022	ROI-based lightweight CNN model	Custom DVS hardware	30 Hz, 0.1°–0.5° accuracy, only 30K parameters
[113]	2022	Query-driven DVS (qDVS) simulation with gaze modeling	Unity-based simulated DVS setup	93.2% horizontal and 93.1% vertical accuracy
[104]	2020	Polynomial regression with hybrid event-frame input	DVS + frame-based camera setup	>10,000 Hz update, 0.45°–1.75° accuracy
[105]	2023	CNN-based pupil tracking from event-to-frame conversion	Event Camera + Pupil Frame Synthesis	3.68 px error, 96% reduction in inference load
[112]	2023	StyleFlow + RetinaFace for RGB-event translation	RGB to Event-translated images	99.4% face, 97.2% pupil accuracy
[109]	2023	Large-scale multimodal fusion model	DAVIS346 Gaze Dataset (1.5M frames, 2.7M refs)	Extensive data for multimodal training
[119]	2023	Multitask model for gaze, head pose, occlusion	Driving dataset with DVS or hybrid input	High-accuracy analytics in real-world scenarios
[106]	2023	Sparse CB-ConvLSTM	Event-based Gaze Dataset (ThreeET)	4.7× fewer operations, maintained accuracy
[115]	2024	End-to-end eye tracking processor	Near-eye DVS Sensor	0.91° accuracy, 1250 Hz, 8.09 mW
[107]	2024	Retina SNN-based neuromorphic model	Custom event-driven pupil tracking setup	3.24 px error, <5 mW consumption
[120]	2024	RNN with coordinate-to-angle loss	Custom event-based gaze dataset	0.46° angular accuracy, 950 Hz update
[121]	2024	SNN vs. ANN for eye tracking	Event camera dataset	87.5% faster inference than ANN
[111]	2024	Dual ResUNet for saccadic gaze prediction	Synthetic Event-based Dataset	99.20% accuracy within 100 px radius
[108]	2024	Wake-up module with adaptive SNN states	Low-power event-based hardware	Improved power savings during idle states
[122]	2024	Causal spatiotemporal ConvNet	AIS 2024 Eye Tracking Dataset	0.9916 p10 accuracy
[114]	2024	Swift-Eye framework with temporal learning	Custom DVS eye tracking benchmark	20% IoU, 12.5% F1 improvement on occlusion
[116]	2024	Bi-GRU with LTV state space module	ThreeET-plus benchmark	SOTA performance on pupil and gaze estimation
[118]	2024	Sparse FPGA Dataflow Accelerator	Custom FPGA + DVS Input	81% p5, 99.5% p10, 0.7 ms latency
[117]	2024	EyeTraES slicing-based adaptive framework	Event-based authentication + gaze dataset	6% IoU improvement, 3× latency reduction

TABLE X
COMPARISON OF PUBLICLY AVAILABLE EVENT-BASED GAZE AND EYE TRACKING DATASETS

Ref.	Sensor Used	Subjects	Events Count	Frame Count	Gaze Tracking Frequency	Accuracy
[104]	DAVIS346	24	Not specified	Not specified	>10,000 Hz	0.45°–1.75°
[106]	Synthetic (v2e)	Not specified	Not specified	Not specified	95 kHz	Not specified
[107]	DVXplorer	30	Variable per recording	Not applicable	≤5 kHz	3-8 px error
[109]	DAVIS346	48	2.7 billion	1.5 million	≤38.4 kHz	1.2-7.7 px
[110]	Prophesee EVK4 HD	66	26.3 billion	3.6 million	≥13 kHz	Not specified

Several studies have tackled the challenges associated with event-based gaze tracking, such as handling abrupt eye movements and occlusions. Table IX shows the summary of work done with event camera for eye gaze and tracking which elaborates the methods/techniques used by the previous works with the sensor/dataset used and the results achieved. A bidirectional long-term sequence modeling approach was introduced to enhance gaze tracking stability by employing a linear time-varying state space module (LTV-SSM), which effectively captures contextual correlations within event sequences [116]. EyeTraES, another advancement, utilizes an adaptive slicing algorithm to optimize event frame accumulation, leading to high pupil tracking fidelity with a 6% improvement in intersection over union (IoU) and reduced latency [117]. In addition, a Sparse FPGA Dataflow Accelerator was developed to efficiently process event-based inputs for low-power gaze tracking, achieving 99.5% accuracy in gaze estimation tasks [118].

A. Datasets

Table X provides a comparative overview of event-based eye-tracking datasets, detailing key attributes such as the sensor

used, the number of subjects, event and frame counts, gaze tracking frequency, and reported accuracy. A dataset collected using the DAVIS346 sensor includes recordings from 24 subjects and is notable for its exceptionally high gaze tracking frequency exceeding 10 000Hz, enabling fine-grained temporal resolution for eye-tracking tasks [104]. Although the total number of events and frames is not specified, the dataset demonstrates strong potential for precision gaze estimation, with reported accuracies ranging from 0.45° to 1.75°. In contrast, a synthetic dataset generated using the v2e (video-to-events) framework stands out for achieving a remarkable gaze tracking frequency of 95 kHz, the highest among current event-based eye-tracking datasets [106]. While this synthetic dataset lacks specific information about subjects and event counts, its frequency performance highlights its value for high-speed gaze tracking research, though its accuracy remains unreported.

The DVXplorer camera was used to create a purely event-driven dataset involving 30 subjects, with variable event counts per recording to accommodate different tracking scenarios [107]. Despite its adaptability, this dataset supports a relatively lower gaze tracking frequency of 5 kHz or less, and it achieves an accuracy between 3 to 8 pixels, which, while not the highest, is suitable for real-time use cases. Another

comprehensive dataset using the DAVIS346 sensor includes 48 subjects and boasts an extensive 2.7 billion recorded events and 1.5 million frames [109]. It supports a tracking frequency of up to 38.4 kHz and reports gaze tracking accuracies ranging from 1.2 to 7.7 pixels, reflecting both its richness and variability across conditions, making it one of the most detailed datasets for real-world event-based gaze estimation.

Another large-scale dataset [110] was recorded using the Prophesee EVK4 HD sensor and includes data from 66 subjects, making it the most diverse in terms of participant representation. It features 26.3 billion recorded events and 3.6 million frames, establishing itself as the most extensive dataset in terms of data volume. With a gaze tracking frequency of at least 13 kHz, it offers a balanced combination of high-speed tracking and large-scale data collection. However, gaze tracking accuracy is not explicitly reported, which limits direct comparisons with other datasets in terms of precision.

B. Challenges

Event-based gaze and eye tracking has shown impressive potential in AR/VR [123], healthcare, and human-computer interaction [124], but several challenges remain before it can be widely adopted. One of the biggest issues is achieving high-frequency tracking with minimal latency. While event cameras can theoretically track eye movements at 10 000+ Hz [104], existing deep learning models struggle to process these data efficiently in real time. The tradeoff between temporal resolution and computational load remains a significant challenge. In addition, event-based gaze tracking datasets are scarce. Existing datasets, such as EV-Eye [109] and RGBE-Gaze [110], are limited in their diversity and fail to capture real-world variations, such as different head poses, occlusions, and lighting conditions. Without sufficient data, models trained on these datasets struggle to generalize across different user demographics and application scenarios. Another major challenge is sensor calibration and cross-modal learning. Many gaze tracking systems use a combination of event cameras, RGB cameras, and infrared sensors, but accurately aligning event data with traditional gaze-tracking methods is nontrivial. Synchronization issues between different modalities can lead to inconsistencies, reducing overall accuracy. Finally, power efficiency remains a concern, particularly for wearable applications such as smart glasses and VR headsets.

VIII. APPLICATIONS OF EVENT CAMERAS IN HRC

Despite their promise in dynamic environments, event cameras remain underexplored in HRC. Although these neuromorphic sensors have shown advantages in tasks like object tracking and motion detection, their potential for enhancing HRC has yet to be fully realized. This gap offers opportunities to leverage event cameras' high temporal resolution and robustness to lighting for advancing HRC systems. Event cameras are transforming HRC by providing high-speed, low-latency, and power-efficient visual perception, allowing robots to understand human actions, gestures, and intentions with unparalleled accuracy.

One of the most crucial applications of event cameras in HRC is real-time human pose estimation, which allows robots

to track human movement continuously without the limitations of frame-based sampling. In industrial environments, where precision and safety are paramount, event cameras provide low-latency skeletal tracking, ensuring that cobots can dynamically respond to worker movements. Unlike RGB cameras, which may struggle with motion blur, event cameras capture even the slightest movement changes, making them ideal for tracking fine motor actions such as tool manipulation, assembly work, or collaborative lifting tasks. In addition, event cameras excel at occlusion handling, meaning that robots can continue to track workers even in crowded workspaces where traditional vision systems might fail.

HAR [125] is another area where event cameras enhance HRC by enabling robots to predict and adapt to human actions in real time. The high temporal resolution of event cameras allows for more accurate classification of activities such as assembling, picking, placing, and quality checking, even in fast-paced environments. For instance, robots can autonomously hand over tools or materials based on a worker's actions, streamlining manufacturing processes. Furthermore, event cameras can monitor ergonomic movements, helping prevent repetitive strain injuries by ensuring that workers follow safe movement patterns. Another critical advantage is anomaly detection, where event cameras can detect sudden deviations, such as a worker slipping or accidentally entering a hazardous zone, prompting robots to take preventive safety measures instantly.

Gesture recognition [126] is another key domain where event cameras offer a fast and energy-efficient alternative to conventional cameras, allowing robots to understand nonverbal commands with minimal processing power. Unlike RGB-based systems that require complex feature extraction to distinguish gestures, event cameras naturally capture the spatiotemporal characteristics of dynamic hand movements with high fidelity. This makes them particularly useful for contactless control in industrial and sterile environments, where workers can instruct robots using simple hand gestures without the need for physical interfaces. In augmented reality (AR) and teleoperation, event cameras enable seamless HRI, allowing operators to remotely guide robotic arms through precise hand and finger movements, even in low-bandwidth scenarios.

In FER, event cameras detect subtle muscle movements at millisecond resolution, overcoming challenges faced by conventional vision systems. This enables emotion-aware robotics that adapt workload or provide personalized assistance, particularly in healthcare and elderly care. Finally, event-based gaze and eye tracking improve intent prediction and cognitive load assessment in HRC by capturing rapid eye movements with minimal latency. Gaze-aware robots optimize task flow by anticipating focus in industrial tasks and facilitate hands-free control in surgery, rehabilitation, and VR/AR applications through ultrafast pupil tracking.

IX. CONCLUSION

Event cameras are changing how robots and humans can work together. Unlike regular cameras, event cameras can see and process movements extremely quickly and work well even

in poor lighting conditions. This review has shown how event cameras are being used in different ways—from tracking human movements and gestures to recognizing facial expressions and following eye movements. These capabilities help robots better understand and respond to humans when working together, making their interactions safer and more efficient. Current research shows that event cameras have some important advantages. They can capture quick movements that regular cameras might miss, and they use less power while doing so. Scientists have developed new ways to process the information from these cameras, leading to better accuracy in recognizing human actions. While there are now more datasets available for testing these systems, researchers still need more real-world examples to make the technology more reliable in everyday situations.

However, there are still some problems to solve before event cameras can be widely used in robots. Researchers need to find better ways to process the unique type of data these cameras produce and combine it with information from other types of sensors. They also need to make sure the systems work well in different environments, like factories or hospitals. As these challenges are addressed, event cameras will become increasingly important in helping robots and humans work together more naturally and effectively.

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